ECHO: THE SIGNAL THAT WAS NEVER SENT

Quantum Investigation Case Report

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1. Final Theory and Verdict

Our provisional theory is that Zara Mahmud's central scientific conclusion was mistaken: the evidence described in the case is consistent with quantum correlations and teleportation, but does not establish that ECHO sent a controllable message faster than light. Standard quantum teleportation requires two ordinary classical bits to reach the receiver before the state can be reconstructed. Entanglement by itself cannot be used to send a chosen message instantaneously.

Confidence: high in the physics conclusion; low in any claim about who caused Zara's disappearance or whether someone deliberately manipulated the records. The participant manual gives motives and conflicting statements for Prof. Imran Vance, Sana Rao, Bilal Noor, and ARIA, but those statements are not proof of guilt. We therefore do not name a culprit on the evidence currently available.

2. Three Important Clues and Evidence Board

FACT (observed / computed)	CLAIM (not independently proven)	INFERENCE (our interpretation)	OPEN QUESTION
The Module 03 Bell-pair circuit predicts only 00/11 when measured in the Z basis; the notebook also tests a changed basis.	Zara said the detector agreement proved a link that could carry a message.	Agreement in one measurement basis alone is insufficient to prove a communication channel. Changed-basis correlations are more informative, but a full Bell-inequality test is stronger evidence.	What measurements did the real detectors make, and were they independently and reliably recorded?
The 32 provided triples decode by majority vote to the text "ECHO," assuming no more than one flip per triple.	The case presents the recovered stream as a damaged copy of Zara's final message.	The coding method can recover this message under the stated error assumption; it does not prove who created or altered the stream.	Was the one-flip-per-triple assumption true for the original machine data?
The Module 06 teleportation circuit includes Alice's two classical measurement bits and receiver corrections.	Zara claimed she teleported a message with no qubit crossing the gap.	Teleportation transfers a quantum state using shared entanglement plus classical communication; it is not faster-than-light signalling.	What exactly did Zara mean by "message," and what happened to the original hardware logs?

3. Experiments and Results

Module 01 — The Gatekeeper (mandatory)

Purpose: check that we can prepare and measure basic qubit states. We used X on $|0\rangle$ to make a certain 1, H on $|0\rangle$ to create a 50/50 superposition, H followed by H to return to $|0\rangle$, and X on qubits 0 and 2 to produce the measured three-bit pattern 101 from 000. Expected ideal results: always 1; approximately half 0 and half 1; always 0 after H-H; and always 101. The H measurement is statistical, so a finite run need not be exactly 50/50. This validates basic circuit behaviour, not the historical events in the lab.

Module 03 — The Two Observers (mandatory)

Purpose: test whether detector agreement alone proves a quantum link. We prepared a Bell pair with H followed by CNOT and measured both qubits. The expected Z-basis outcomes are 00 and 11, with no 01 or 10 in the ideal circuit. We then applied H to both qubits before measurement to compare correlations in the X basis with a classical shared-bit mixture. This illustrates why a single matching result is not enough. The simplified comparison is not a substitute for a full Bell-inequality experiment against all local hidden-variable models.

Module 05 — The Damaged Message (mandatory)

Purpose: recover a message from repeated, noisy bits. We grouped the 96-bit stream into 32 triples and used majority vote for each triple. The decoded bytes spell "ECHO." This works if at most one bit in each triple was flipped. If two copies flip—for example, original 000 received as 011—the majority becomes wrong. The three-bit repetition code can correct one flip per triple; it cannot always recover the original when two bits flip. An unknown qubit also cannot simply be copied three times because of the no-cloning theorem.

Module 06 — The Impossible Claim (stretch)

Purpose: test the faster-than-light interpretation. The circuit prepares an input state, creates an entangled pair, performs Alice's measurements, sends two classical bits, and applies conditional corrections at Bob's end. The expected result is that Bob's corrected qubit reproduces the input state statistically. If the classical bits are unavailable, Bob cannot know which correction to apply, so the shared entanglement alone does not deliver a usable chosen message. In the notebook the input state uses $\theta = \pi/3$, so we predicted $P(1) = 0.25$ for Bob when the corrections are applied, and $P(1) = 0.50$ (a maximally mixed qubit, i.e. no message) when the classical bits are withheld. This supports the standard no-faster-than-light conclusion.

Results:

Circuit diagrams, measurement counts and histograms for every experiment are in the notebook. The deterministic circuits (X, H–H, the 101 vault, and the Bell pair in the Z basis) are predicted to give exactly one outcome or only the outcomes listed above; the H circuit and the teleportation runs are statistical, so small deviations from 50/50 and 0.25 are expected.

4. Interpretation and Limitations

Together, the experiments support a cautious interpretation: quantum systems can show correlations that a simple classical shared-bit model does not reproduce under the tested basis changes, and quantum teleportation still needs a classical channel. These results challenge Zara's faster-than-light messaging

claim. They do not prove that the original ECHO hardware behaved exactly like our simulator, that every log is authentic, or that any named person caused Zara's disappearance. The notebook reconstructs the manual's tasks; it does not inspect the original machine or recover hidden hardware logs.

5. Alternative Explanation and Next Experiment

Strongest alternative: The experiment may have been technically legitimate but misinterpreted, while the missing records and Zara's disappearance are unrelated or have an ordinary explanation. A different possibility is deliberate log or configuration tampering, but the manual's character statements alone do not distinguish these accounts.

Next experiment: Repeat the detector test with independently selected measurement settings on both sides and collect enough trials to evaluate a properly designed Bell-inequality test. If the data violate the relevant Bell inequality with suitable controls, that would rule out a broad class of local hidden-variable explanations for the correlations; it still would not demonstrate faster-than-light messaging. Separately, compare signed backups, access records, and timestamps to investigate possible log tampering. Authentic, consistent records would weaken the tampering theory; verifiable inconsistencies would strengthen the case for it.

6. References and Assistance

Primary source: ECHO Participant Manual, "ECHO: The Signal That Was Never Sent," case file QX-01 and modules 01, 03, 05, and 06. Quantum circuit simulations: Qiskit and Qiskit Aer, as used in the accompanying notebook. AI assistance: ChatGPT was used to help draft an early version of this report and to explain circuit concepts. Claude (Anthropic) was used to review the notebook and report against the submission guidelines, tidy the notebook code, and suggest wording changes. I reviewed the final notebook and report and am responsible for the conclusions.